COMPUTATION OF THE TWISTED INVERSE IMAGE FOR AN EFFECTIVE CARTIER DIVISOR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The reduction of the assertion to a statement about the internal hom of the algebra into the ring. Invoked inside the counted proof (source lines 2632-2635). Reduces to computing one object in the derived category.
- The compatibility of the internal hom with the restriction functor. Invoked inside the counted proof (source lines 2641-2644). Allows the computation to be carried out over the ambient ring.

Source reference. Original lemma lemma-RHom-ext: sources/stacks-dualizing/pinned-original-source.tex, source lines 2452–2452. Navigation pointer only: the counted text refers to the Stacks item lemma-RHom-ext, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-RHom-is-tensor-special: sources/stacks-dualizing/pinned-original-source.tex, source lines 2582–2582. Navigation pointer only: the counted text refers to the Stacks item lemma-RHom-is-tensor-special, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $R \to A$ be a surjective ring map whose kernel I is an invertible R-module. The functor $R\mathrm{Hom}(A, -) : D(R) \to D(A)$ is isomorphic to $K \mapsto K \otimes^{\mathbf{L}}_R N[-1]$ where N is inverse of the invertible A-module $I \otimes_R A$.*

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Proof. Since A has the finite projective resolution

$$0 \to I \to R \to A \to 0$$

we see that A is a perfect R-module. By Lemma lemma-RHom-is-tensor-special it suffices to prove that $R\mathrm{Hom}(A, R)$ is represented by $N[-1]$ in $D(A)$. This means $R\mathrm{Hom}(A, R)$ has a unique nonzero cohomology module, namely N in degree 1. As $\mathrm{Mod}_A \to \mathrm{Mod}_R$ is fully faithful it suffice to prove this after applying the restriction functor $i_* : D(A) \to D(R)$. By Lemma lemma-RHom-ext we have

$$i_* R\mathrm{Hom}(A, R) = R\mathrm{Hom}_R(A, R)$$

Using the finite projective resolution above we find that the latter is represented by the complex $R \to I^{\otimes -1}$ with R in degree 0. The map $R \to I^{\otimes -1}$ is injective and the cokernel is N. $\square$